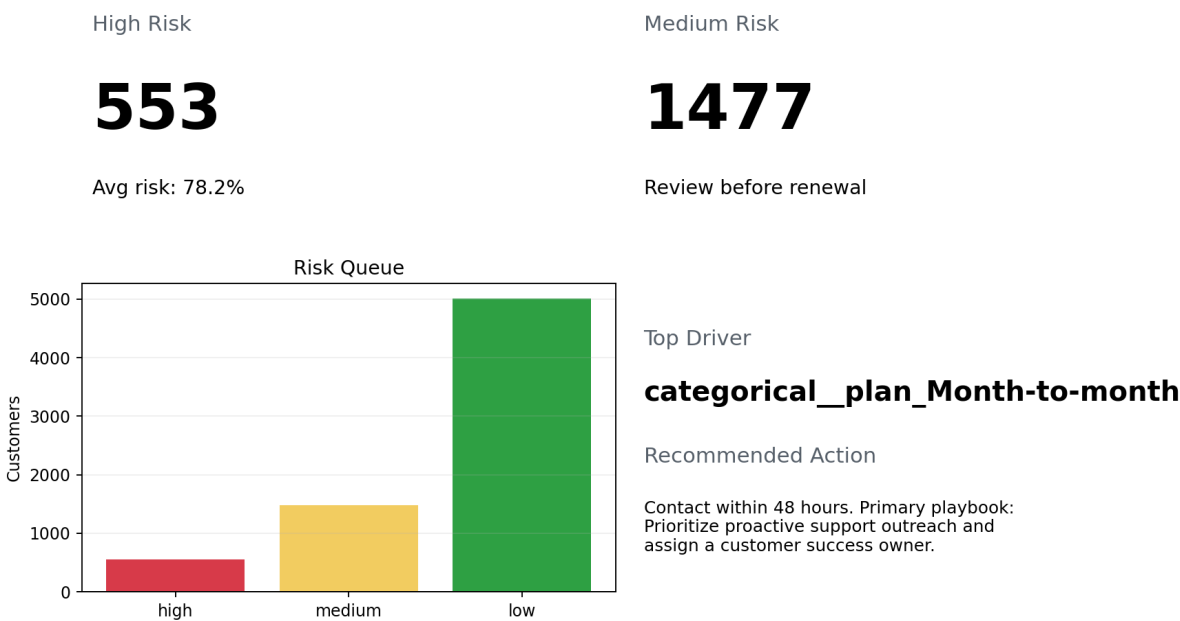


# Churn Prediction + Retention Report

Executive summary: this report ranks customers by churn risk and turns model output into a prioritized retention action queue.

The model ranks customers by churn risk. The top-risk segment should be reviewed first because it concentrates the highest expected churn probability.

## Churn Retention Executive Snapshot



## Dataset Summary

| Metric                 | Value  |
|------------------------|--------|
| rows                   | 7043   |
| columns                | 11     |
| observed_churn_rate    | 26.54% |
| duplicate_customer_ids | 0      |
| leakage_warnings       | 0      |

## Model Metrics

| Metric          | Value  |
|-----------------|--------|
| roc_auc         | 0.8435 |
| pr_auc          | 0.6564 |
| baseline_pr_auc | 0.2655 |
| accuracy        | 0.7908 |

| Metric           | Value                     |
|------------------|---------------------------|
| precision        | 0.6319                    |
| recall           | 0.508                     |
| f1               | 0.5632                    |
| confusion_matrix | [[1386, 166], [276, 285]] |

## Risk Segment Breakdown

| risk_segment | customers | avg_churn_probability |
|--------------|-----------|-----------------------|
| high         | 553       | 0.7818                |
| medium       | 1477      | 0.5405                |
| low          | 5013      | 0.1267                |

## Lift Table Interpretation

The highest-risk 10% captures 28.9% of observed churners. The highest-risk 20% captures 48.1%. Review the 553 high-risk customers first.

| decile | customers | churners | churn_rate | avg_churn_probability | cumulative_churners | cumulative_churn_rate |
|--------|-----------|----------|------------|-----------------------|---------------------|-----------------------|
| 1      | 212       | 162      | 0.7642     | 0.7496                | 162                 | 0.2888                |
| 2      | 211       | 108      | 0.5118     | 0.5772                | 270                 | 0.4813                |
| 3      | 211       | 100      | 0.4739     | 0.4521                | 370                 | 0.6595                |
| 4      | 211       | 77       | 0.3649     | 0.3273                | 447                 | 0.7968                |
| 5      | 212       | 43       | 0.2028     | 0.2296                | 490                 | 0.8734                |

## Top Recommended Actions

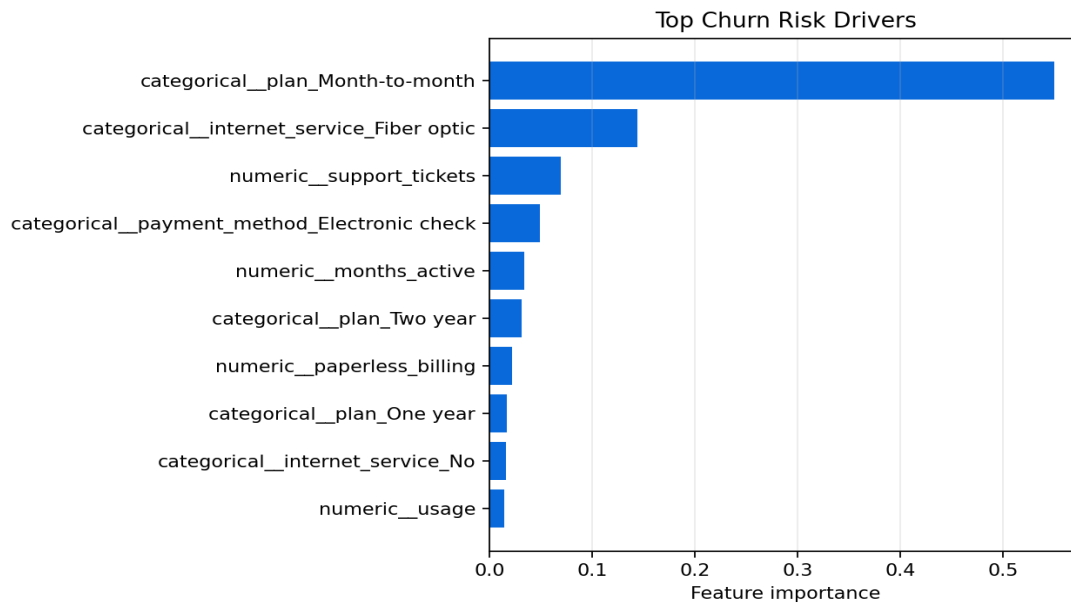
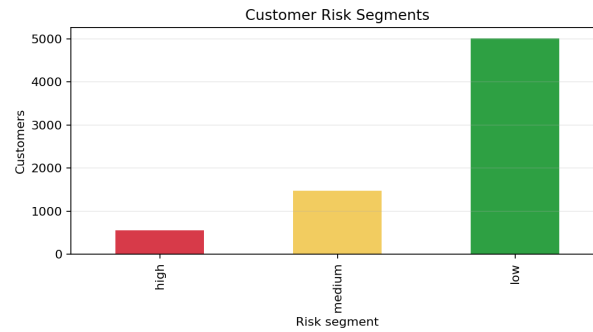
| customer_id | risk_segment | churn_probability | likely_drivers            | retention_priority |
|-------------|--------------|-------------------|---------------------------|--------------------|
| 7216-EWTRS  | high         | 0.9061            | support_tickets, payments | 1                  |
| 6567-HOOPW  | high         | 0.9058            | support_tickets, payments | 1                  |
| 8580-AECUZ  | high         | 0.9058            | support_tickets, payments | 1                  |
| 9300-AGZNL  | high         | 0.9054            | support_tickets, payments | 1                  |
| 9497-QCMMS  | high         | 0.9040            | support_tickets, payments | 1                  |

Detailed recommended\_action text is delivered in retention\_recommendations.csv so the PDF stays readable.

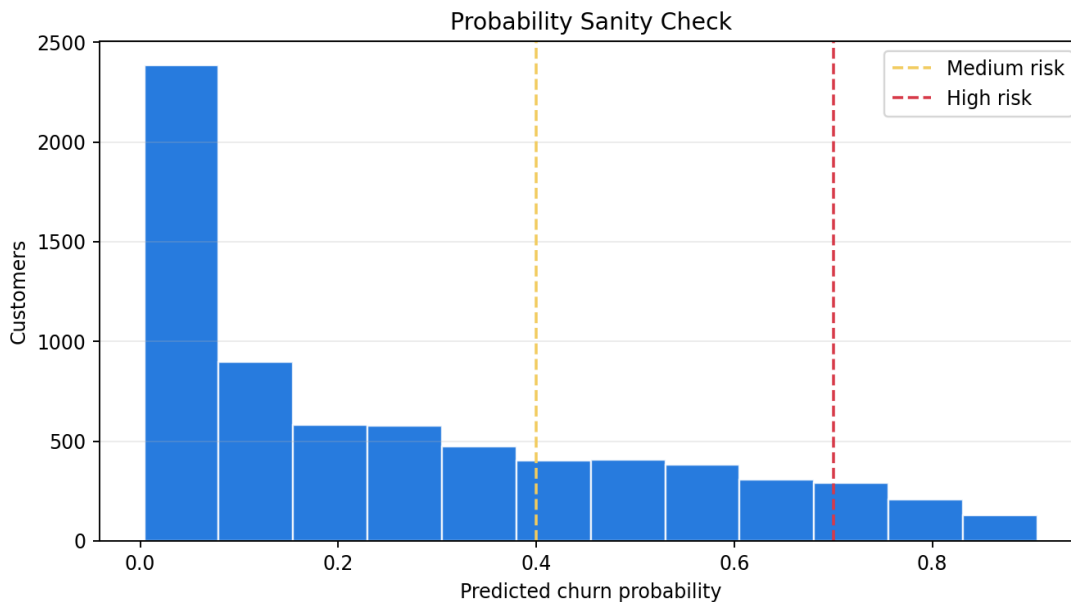
## Recommended Retention Strategy

Use the retention recommendation CSV as the operating queue. High-risk customers should receive direct outreach; medium-risk customers should receive targeted education, usage review, or plan-fit checks.

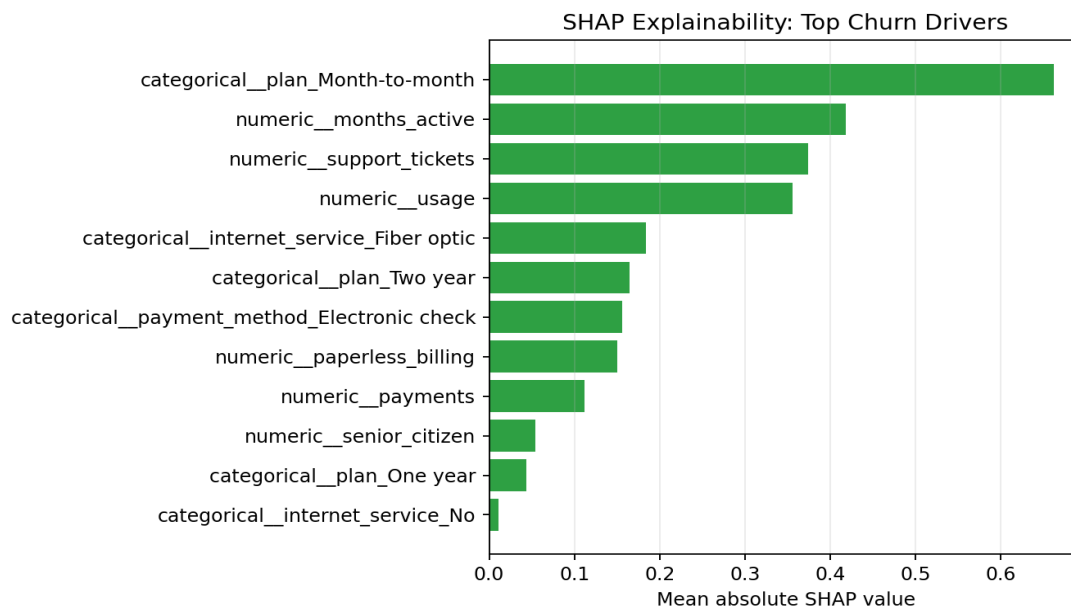
## Visual Evidence



## Probability Sanity Check



## SHAP Explainability



## Limitations

This report does not guarantee a specific accuracy number. Results depend on data volume, feature quality, label quality, and whether post-churn leakage columns were excluded before modeling.

## Next Steps

Recommended next steps: review leakage warnings, confirm the business definition of churn, enrich weak features, run the action queue, and compare retention outcomes after the next billing cycle.

## Scope and Privacy

This repository keeps raw private datasets out of version control. The checked-in sample artifacts are reproducible from the included public-data workflow and are meant to show the report shape without exposing private data.